

CLoRA: Contrastive Test-Time Composition of Multiple LoRA Models

Replication + Improvement Project | CS683 Generative AI

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Project Snapshot



Problem we address

Multi-LoRA composition should place multiple personalized concepts or styles in one image, but direct merging often makes one concept dominate.

Core idea

Multi-branch denoising + contrastive attention update + attention-mask fusion.

Project setup

Project type	Replicate + Improve
Base model	Stable Diffusion v1.5
Platform	Kaggle, 2× Tesla T4 GPUs
Main baseline	Naive LoRA Merge

Final claim

We reproduced the key trend of the paper:
CLoRA improves over naive LoRA Merge on every evaluated metric in our controlled setup.

Problem: Why Multi-LoRA Merge Fails

- LoRAs are lightweight adapters for a subject, concept, or visual style.
- A useful goal is to combine two LoRAs in the same generated image.
- Naive merging activates both adapters globally, so the model has no clear region ownership.

Observed failures

Attention overlap: both LoRAs focus on the same area.

Attribute binding: style or identity features leak across concepts.



Our LoRA Merge baseline: visually plausible images, but weak



Reference Method: CLoRA in Three Steps



1. Multi-branch denoising

Run the base branch and one LoRA branch per concept at each diffusion step. All branches share the same latent z_t .

2. Contrastive update

Group attention maps by concept. Same-concept maps are pulled together; different concepts are pushed apart using InfoNCE.

3. Masked fusion

Convert attention maps into masks, so each LoRA affects only its relevant spatial region.

Latent update

$$z'_t = z_t - \alpha_t \nabla_{z_t} \mathcal{L}_{\text{InfoNCE}}$$

Mask construction

$$M[x, y] = \mathbb{1}(A[x, y] \geq \lambda \max A)$$

Our Implementation vs. Paper

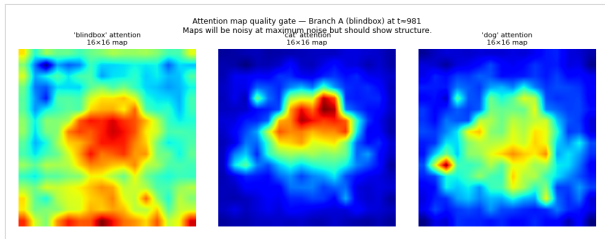


Component	Paper	Our implementation
Hardware	V100 / H100; about 25 GB for two LoRAs	Kaggle 2× Tesla T4, 31.2 GB total
Base model	Stable Diffusion v1.5	Stable Diffusion v1.5
Denoising	50 steps, DDIM-style sampling	50 steps, DDIM-style sampling
Loss settings	InfoNCE, $\tau = 0.5$	InfoNCE, $\tau = 0.5$
Mask threshold	$\lambda = 0.3$	$\lambda = 0.3$
Update schedule	Steps $\{0, 10, 20\}$, cutoff after step 25	Same schedule
LoRA collection	Large subject/character benchmark	Community LoRAs: blindbox, moxin, colorwater, fantchar
Evaluation scale	131 LoRAs, 200 prompts, 10 seeds	Smaller controlled benchmark, 5 seeds

Main implementation work

Adapter switching, custom cross-attention extraction, InfoNCE latent update, attention-mask fusion, DINO/CLIP evaluation, and SDXL masked-fusion extension.

Attention Map Verification



Cross-attention quality gate from our implementation.

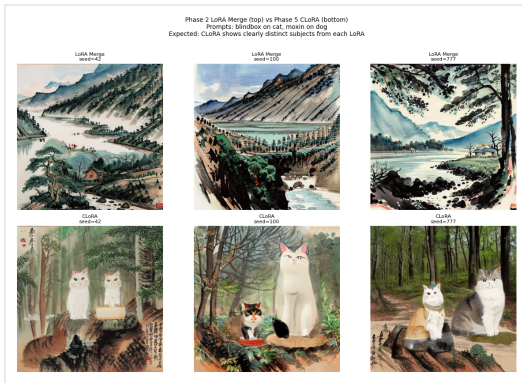
Check	Value
Row-sum deviation	3.78×10^{-4}
Cat peakedness	3.40
Dog peakedness	2.62
Cat-dog overlap	0.643
Threshold	0.85

Takeaway

Attention maps are not blank or random. Different tokens show different spatial focus, so the contrastive objective has meaningful signal.



Qualitative Result: Merge vs. CLoRA



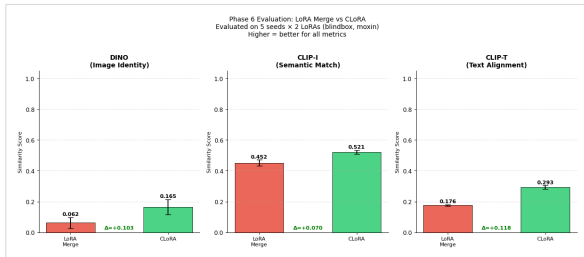
LoRA Merge

One style usually dominates and the intended multi-concept control is weak.

CLoRA

Better spatial separation and stronger composition than direct merging.

Quantitative Result: CLoRA Beats Merge



Metric	Merge	CLoRA	Gain
DINO Avg	0.062	0.165	+166%
CLIP-I Avg	0.452	0.521	+15%
CLIP-T	0.176	0.293	+66%

Interpretation

Absolute values are lower than the paper because our setup is smaller and uses mostly style LoRAs. The important trend is identical: **CLoRA improves over LoRA Merge**.

DINO measures visual similarity, CLIP-I measures image-image semantic similarity, and CLIP-T measures text alignment.

Replication Scorecard



Verified

- Training-free pipeline
- SD1.5 base model
- Multi-branch denoising
- 16×16 attention maps
- InfoNCE update
- Attention masks
- CLoRA $>$ Merge on all metrics

Approximated

- Smaller LoRA collection
- Mostly style LoRAs
- 5 seeds instead of full paper scale
- Internal comparison, not full benchmark reproduction

Not done

- 131-LoRA benchmark
- 50-participant user study
- ZipLoRA / Mix-of-Show reruns
- Full SDXL InfoNCE backprop

Honest summary

We replicated the algorithmic pipeline and confirmed the same improvement direction, but we do not claim an exact reproduction of the paper's full-scale benchmark.

Our Improvement: SDXL Masked-Fusion Extension



Phase 7 — SD1.5 512×512 (top) vs SDXL 768×768 (bottom)
3d render style → cat | Ikea instructions → dog



Property	SD1.5	SDXL
Output	512×512	768×768
Attention grid	16×16	24×24
Spatial tokens	256	576
UNet size	860M	2.57B
Text encoders	1	2

What worked

Masked fusion was ported to SDXL, giving sharper 768×768 outputs.

Limit

Full SDXL InfoNCE backprop was skipped because it exceeded T4 VRAM.

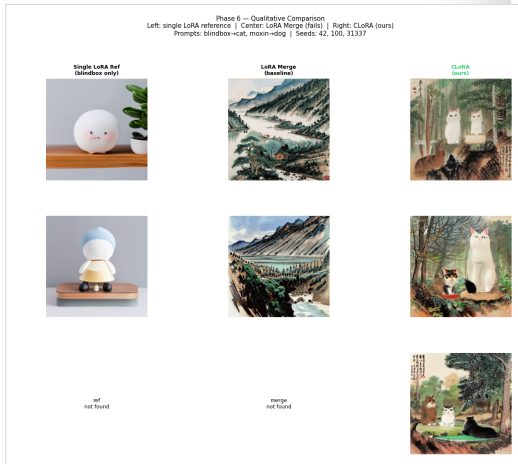
Engineering Contributions on Kaggle/T4

Practical fixes

1. Disk overflow during downloads
2. Broken bitsandbytes / triton import
3. Blank outputs from checkpointing during inference
4. SDXL VAE fp16 NaN / black-image issue
5. Cross-device DDIM arithmetic issue

Why it matters

These fixes made multi-branch LoRA diffusion possible on limited Kaggle T4 hardware.



Qualitative comparison from our final pipeline.



Conclusion and Future Work



What we achieved

- Implemented CLoRA core pipeline from scratch.
- Verified attention extraction with quality gates.
- Applied InfoNCE updates and masked fusion.
- Improved over LoRA Merge on all metrics.
- Built an SDXL masked-fusion extension.

Future work

- Run full SDXL InfoNCE on A100/H100.
- Use subject LoRAs closer to the paper benchmark.
- Scale to three or more LoRAs.
- Add automatic token-to-mask selection.
- Evaluate with a larger reference bank.

Thank You